SEC 4 · LEC 2 · CHEMICAL ENGINES (ROCKETS)

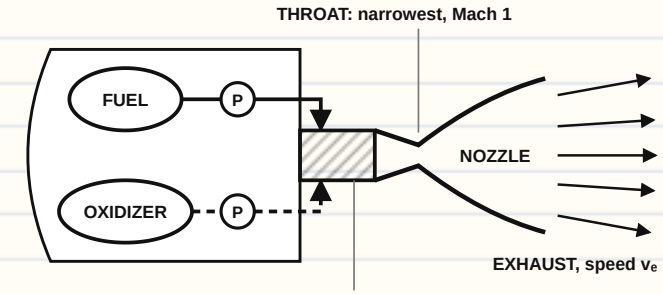
Fire, a squeeze, and a bell

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S4 / L2 — Chemical engines **date · pg __**

1 · LIQUID-PROPELLANT CHEMICAL ENGINE



2 · THE NOZZLE TURNS PRESSURE INTO SPEED

CHAMBER	→	THROAT	→	BELL EXIT
pressure HIGH speed LOW		squeezed to the speed of sound		pressure LOW speed 3–4 km/s

hot + squeezed gas IN → fast gas OUT

3 · THE TWO NUMBERS

SPECIFIC IMPULSE	THRUST
$I_{sp} = v_e \div g_0$	$F = \dot{m} \times v_e$
seconds · the mileage $g_0 = 9.81\ m/s^2$ 3,414 m/s → 348 s	newtons · the push $\dot{m}$ = kg thrown per second 287 × 3,414 ≈ 981,000 N

chemical: Isp LOW · $\dot{m}$ HUGE → thrust HUGE

✏️ **PENCIL KEY** · solid pipe = fuel · dashed pipe = oxidizer · hatched box = where it burns · thin arrows = exhaust · Ⓟ = pump

⭐ **3 TAKEAWAYS** 3 lines

1. Burn fuel + oxidizer, squeeze the gas through a throat, the bell makes it fast.
2. Isp = $v_e \div g_0$ is the mileage (s). $F = \dot{m} \times v_e$ is the push (N).
3. Chemical: low Isp, but tonnes thrown per second, so huge thrust.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What is a chemical rocket in one line?	chemical engine = reaction engine: stored propellant → high-speed jet (Newton 3)
Why carry an oxidizer?	propellant = fuel + oxidizer · no air in space → bring your own oxygen · e.g. liquid hydrogen + liquid oxygen
Path of the propellant, tank to exhaust?	tanks → pumps → combustion chamber (burns, ~3,300 K) → throat → nozzle
What does the nozzle do? Where is the gas fastest?	nozzle: pressure + heat → speed · bell shape lets the gas expand
Isp: formula, unit, what it measures?	Isp = $v_e \div g_0$ · g_0 = 9.81 m/s² · unit: seconds · = efficiency · higher v_e → less propellant for the same push
Thrust: formula? Two ways to raise it?	thrust $F = \dot{m} \times v_e$ · $\dot{m}$ = mass flow, kg per second · unit: newtons
"Low Isp but high thrust." How can both be true?	chemical: Isp low (250–450 s) but $\dot{m}$ huge (tonnes/s) → thrust huge
Solid vs liquid?	best thrust for the engine's weight · the propellant is the heavy part (~90% of a rocket)
? what do the two engines in the photos burn	solid rocket: fuel + oxidizer premixed in one block · can't shut off · often boosters
	e.g. SpaceX 2nd stage (1 engine) · Saturn V (5 engines) · both kerosene + liquid oxygen

SEC 4 · LEC 2 · TEACH

A fire with only one way out

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Blow up a balloon and let it go. Air squirts out the neck one way and the balloon shoots off the other way. A rocket engine is a balloon that keeps refilling itself by lighting a fire inside. A fire needs two things: something to burn (**fuel**) and oxygen. There is no air in space, so the rocket carries its own oxygen in a second tank (**oxidizer**).

The fire makes gas that is very hot and very squeezed, and it has only one way out: a small hole called the **throat**, then a bell called the **nozzle**. The bell is where the magic happens. It lets the squeezed gas spread out, and as it spreads it speeds up, to about four times the speed of a rifle bullet. Two numbers describe any engine: how hard it pushes (**thrust**) and how far it gets on each kilo of propellant (**specific impulse**). A chemical rocket is a drag racer: enormous push, terrible mileage.

🏷 The official names (name first, then what it means)

- **Chemical rocket engine** (chemical propulsion): makes thrust by burning propellant and expelling the gas.
- **Bipropellant** (biprop): an engine with two propellants, a **fuel** and an **oxidizer**. The slide's diagram is one. A **monopropellant** engine uses a single liquid that breaks down on a catalyst (hydrazine).
- **Cryogenic propellants**: liquids kept very cold. **LH_2** (liquid hydrogen, −253 °C) and **LOX** (liquid oxygen, −183 °C). **RP-1** is refined kerosene, stored at room temperature.
- **Turbopump** (the slide's "pressure pump"): a turbine-driven pump that forces propellant into the chamber. Small engines skip it and push with tank pressure: **pressure-fed**.
- **Combustion chamber** (thrust chamber): where the propellants mix and burn.
- **de Laval nozzle** (converging–diverging nozzle, bell nozzle): narrows to a **throat**, then widens. Turns pressure and heat into exhaust speed.
- **Exhaust velocity (v_e)** (the instructor's "velocity of ejection"): how fast the gas leaves.
- **Mass flow rate ($\dot{m}$)**, said "m-dot" (his "flux of propellant"): kilograms leaving per second.
- **Specific impulse (Isp)** = $v_e \div g_0$, in seconds. **g_0** is **standard gravity**, 9.80665 m/s².
- **Solid rocket motor** (SRM): fuel and oxidizer cast together as one solid block, the **grain**.
- **Thrust-to-weight ratio**: an engine's thrust divided by its own weight.

🛠 How it actually works

Step	The mechanism, with numbers
1 · Two tanks	Burning is fuel joining with oxygen. Hydrogen + oxygen gives steam and 13 million joules per kg of mix. A jet plane takes its oxygen from the air; a rocket must carry it, and the oxidizer is most of the load (about 6 kg of oxygen per kg of hydrogen).
2 · Feed	The chamber is at 100+ times air pressure, so propellant has to be forced in. Each Saturn V F-1 engine had a 41 MW turbopump moving **2.6 tonnes every second**.
3 · Burn	In the chamber: about 3,300 K (hotter than steel's melting point of 1,700 K) and 100–300 bar. The gas is hot and squeezed but slow.
4 · Throat	The narrowing speeds the gas up until, at the narrowest point, it hits exactly the speed of sound (Mach 1).
5 · Bell	Past Mach 1 the rule flips: a *widening* pipe speeds the gas up. Pressure and temperature fall, speed climbs. SpaceX's Merlin Vacuum exits at **3.4 km/s**.
6 · Thrust	$F = \dot{m} \times v_e$ = 287 kg/s × 3,414 m/s ≈ **981,000 N**, enough to hold up 100 tonnes.
7 · Isp	Isp = 3,414 ÷ 9.81 = **348 s**. In words: 1 kg of propellant can give 9.81 N of push (1 kg of weight) for 348 seconds.
8 · The ceiling	The energy in a kilo of propellant is fixed by chemistry. Even at 100% conversion, 13 MJ/kg gives $v_e = \sqrt{(2 \times 13{,}000{,}000)} \approx 5.2$ km/s. Real best: 4.5 km/s, Isp about 460 s. No chemical engine will ever reach 1,000 s.
9 · Not melting	The cold fuel is piped through channels in the chamber and nozzle walls before it burns: **regenerative cooling**.

SEC 4 · LEC 2 · NUMBERS + FACT CHECK

Real engines, real numbers

Check $F = \dot{m} \times v_e$ and $Isp = v_e \div g_0$ against real engines, then the caption errors and where the instructor was loose or wrong.

🚀 Engines you can check the formulas on

Engine	Where	Propellants	Thrust	Isp	$v_e = Isp \times 9.81$	$\dot{m} = F \div v_e$
F-1	Saturn V stage 1, ×5 (right photo)	kerosene (RP-1) + LOX	6,770,000 N	263 s	2.6 km/s	2,600 kg/s each
Merlin Vacuum	Falcon 9 stage 2, ×1 (left photo)	kerosene (RP-1) + LOX	981,000 N	348 s	3.4 km/s	287 kg/s
RS-25	Space Shuttle, SLS	LH_2 + LOX	2,279,000 N	452 s	4.4 km/s	514 kg/s
Solid booster	Space Shuttle, ×2	solid grain	12,500,000 N	268 s	2.6 km/s	4,700 kg/s
Hydrazine thruster	most spacecraft	hydrazine	1–22 N	225 s	2.2 km/s	grams per second

Hydrogen gets the best mileage; kerosene and solids the worst, yet they give the biggest push. All figures rounded; F-1 and booster thrust at liftoff, the rest in vacuum.

🔎 Caption errors

Captions said	Instructor meant
"chemical IntoNow heat engines"	chemical **internal** heat engines
"they have no specific impulse"	**low** specific impulse
"velocity of a section / rejection"	velocity of **ejection** (exhaust velocity)
"divided by zero"	divided by **g-zero (g_0)**
"nine point ninety nine point eighty eight to one"	**9.81** m/s^2
"specific forces / specific examples"	specific **impulse**
"threat / frost / thirst"	**thrust**
"expecting a lot of mass"	**expelling** a lot of mass
"thermal specific properties… higher food"	**thermodynamic** properties… **fluid**
"in this cost"	in this **course**

📒 Question bank cards

- Trace a kilo of propellant from tank to exhaust: five stops.
- Where in the engine is pressure highest? Where is speed highest?
- Isp 300 s. What is the exhaust velocity? What does "300 seconds" mean?
- An engine throws 500 kg/s at 4,400 m/s. Thrust?
- Why can't a chemical engine reach Isp 1,000 s?
- Chamber gas is hotter than steel melts. Why doesn't the engine melt?

▲ Where the instructor was loose

- **"They are the lightest and have the highest thrust."** The engines have the best thrust for their own weight (a Merlin lifts about 180 times its weight). The rocket as a whole is the heaviest option, because about 90% of it is propellant.
- **"The nozzle shape maximizes the pressure we get out."** Backwards. The nozzle *spends* pressure to buy speed. Pressure is highest in the chamber and lowest at the exit.
- **"It's quite common that fuel is liquid hydrogen."** Common on upper stages. But both engines in his photos burn **kerosene** with liquid oxygen.
- **"In the United States you may use different units."** That is the whole reason Isp is in seconds: dividing by g_0 gives the same number in metric and imperial.
- **"Solid rockets, usually known as boosters."** Solid is a propellant type; booster is a job. Solids also serve as final kick stages (New Horizons left Earth on one), and boosters can be liquid.
- **"Pressure pump" on the slide** merges two feed systems: a turbopump, or no pump and pressurized tanks.
- **Not said: what spacecraft themselves carry.** Hydrogen and oxygen boil away in days. Probes use storable propellants that can sit in tanks for 20 years: hydrazine alone, or hydrazine-family fuel with nitrogen tetroxide (Isp about 320 s). They ignite on contact, with no spark: **hypergolic**.
- **Not said: why the bell is so big on the second stage.** In vacuum the gas can keep expanding, so a longer bell extracts more speed.

SEC 4 · LEC 2 · VOCABULARY

Every word used, in plain English

46 entries: the lecture's words plus every term I used on the other pages. "Repeat" = you have met it before.

Chemical engine / chemical rocket — an engine that makes thrust by burning propellant and throwing out the hot gas. *Balloon with a fire in it.*

Internal heat engine — the instructor's phrase: a machine that burns fuel inside itself and turns the heat into motion.

Stored mass / propellant — everything the rocket carries to throw out. Fuel plus oxidizer. *Repeat.*

Propulsive jet — the stream of fast gas leaving the nozzle.

Reaction engine — any engine that works by throwing mass backward (Newton's third law). *Repeat, L1.*

Thrust (F) — the engine's push, in newtons. $F = \dot{m} \times v_e$. *Repeat, L1.*

Overall thrust — just 'total thrust'. Nothing new.

Specific impulse (Isp) — exhaust velocity divided by g_0, in seconds. Push per kilo of propellant. *Miles per gallon.*

Exhaust velocity (v_e) = velocity of ejection — how fast the gas leaves the nozzle, relative to the rocket.

g_0 = standard gravity — 9.80665 m/s², the pull of gravity at Earth's surface, used as a fixed constant. *A unit converter, nothing more.*

Constant — a number that never changes. g_0 is one.

Efficiency — how much useful output you get per unit of input. For engines: Δv per kilo of propellant.

Mass flow rate ($\dot{m}$) = flux of propellant — kilograms of propellant leaving per second. Said 'm-dot'. *Repeat, L1.*

Force / newton (N) — a push or pull / its unit. kN = thousand, MN = million. *Repeat, L1.*

Fuel — the part of the propellant that burns: hydrogen, kerosene, hydrazine.

Oxidizer (oxidiser) — the part that supplies the oxygen: liquid oxygen, nitrogen tetroxide. *The air you brought with you.*

Bipropellant / monopropellant — two propellants that burn together / one liquid that breaks down by itself on a catalyst.

Liquid hydrogen (LH_2) — hydrogen chilled to −253 °C. Best mileage, very bulky, boils away.

Liquid oxygen (LOX) — oxygen chilled to −183 °C. The usual oxidizer for launch rockets.

Cryogenic — stored extremely cold to stay liquid.

Kerosene / RP-1 — refined jet-fuel-like liquid. Dense, room temperature, lower Isp than hydrogen.

Hypergolic / storable — propellants that ignite on contact, and keep for years in a tank. *What probes carry.*

Hydrazine — the standard storable spacecraft propellant. Toxic. Isp about 225 s alone.

Pump / turbopump — forces propellant into the high-pressure chamber. Driven by a small turbine.

Pressure-fed — no pump: high-pressure gas in the tanks pushes the propellant out. Simple, for small engines.

Combustion / to burn — fuel joining with oxygen fast, releasing heat.

Combustion chamber — the strong can where the burning happens.

Reaction (chemical) — here: the burning itself. Not the same 'reaction' as in action–reaction.

Thermodynamic properties — the state of a gas: its temperature, pressure and density.

Fluid — anything that flows: a liquid or a gas.

Pressure / bar — how hard a gas pushes on its walls. 1 bar ≈ normal air pressure. *Repeat.*

Throat — the narrowest point of the nozzle. Gas reaches the speed of sound there.

Nozzle — the shaped outlet that turns pressure and heat into speed. *The bell.*

de Laval nozzle = converging-diverging nozzle = bell nozzle — narrows, then widens. Named after Gustaf de Laval, 1888. *Add to Who's Who.*

Expand / expansion — gas spreading into more space. Its pressure and temperature drop.

Parabolic — the curved bell shape; a practical approximation of the ideal contour.

Speed of sound / Mach 1 / supersonic — how fast pressure waves travel in the gas / moving at that speed / faster than it.

Kinetic energy — energy of motion. The nozzle turns heat energy into it. *Repeat.*

Regenerative cooling — running cold fuel through the engine walls to keep them from melting.

Solid rocket motor / grain — fuel and oxidizer cast as one rubbery block / that block. Lights once, can't be shut off.

Booster — a rocket strapped on to help at liftoff. A job, not a propellant type.

Stage / second stage / staging — a rocket section with its own engines and tanks, dropped when empty. *Repeat idea: rocket equation.*

Thrust-to-weight ratio — thrust divided by weight. Above 1 means it can lift itself.

SpaceX / Falcon 9 / Merlin Vacuum — US rocket company / its workhorse rocket / that rocket's second-stage engine. *Add to Who's Who.*

Saturn V / F-1 — the Apollo Moon rocket (1967–1973) / its first-stage engine, the most powerful single-chamber liquid engine flown. *Add to Who's Who.*

RS-25 — the Space Shuttle main engine, hydrogen and oxygen. *Add to Who's Who.*

SEC 4 · LEC 2 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D for this lecture. The concept cards (route E) are on the next page.

A · Equations & key numbers page (p3–4)

$Isp = v_e \div g_0$	$g_0 = 9.81\ m/s^2$ · unit: seconds
	3,414 m/s ÷ 9.81 = 348 s (Merlin Vacuum)
$v_e = Isp \times g_0$	452 s × 9.81 = 4.4 km/s (hydrogen engine)
$F = \dot{m} \times v_e$	287 kg/s × 3,414 m/s ≈ 981,000 N
chemical ceiling	$v_e \approx 4.5$ km/s · Isp ≈ 460 s
chamber	~3,300 K · 100–300 bar · throat = Mach 1

B · Section 4 map page (add to the sentence from L1)

chemical: fuel + oxidizer burn → throat (Mach 1) → bell turns pressure into speed; low Isp, huge $\dot{m}$ → huge thrust

D · Question bank ("but how?" cards)

Q: How does a nozzle make gas go faster?
A: It spends pressure and heat to buy speed. Throat = Mach 1, then the widening bell speeds supersonic gas up further.
Q: What does "Isp = 348 s" physically mean?
A: 1 kg of propellant can give 9.81 N of thrust for 348 s. Same as v_e = 348 × 9.81 = 3.4 km/s.
Q: Low Isp but high thrust. How?
A: $F = \dot{m} \times v_e$. v_e is modest but $\dot{m}$ is tonnes per second.
Q: Why doesn't the chamber melt at 3,300 K?
A: Regenerative cooling: cold fuel runs through the walls first.

C · One-line word bank (new lines)

p__ · chemical rocket · burn propellant, throw the gas
p__ · fuel / oxidizer · what burns / the oxygen you bring
p__ · bipropellant / monopropellant · two liquids burn / one breaks down
p__ · LH_2 / LOX = liquid hydrogen / oxygen · cryogenic, best Isp
p__ · RP-1 = kerosene · dense, room temp, lower Isp
p__ · hypergolic / storable · ignite on contact, keep for years
p__ · hydrazine · spacecraft's standard propellant, ~225 s
p__ · turbopump / pressure-fed · pump forces it in / tank gas does
p__ · combustion chamber · where it burns, ~3,300 K
p__ · throat · narrowest point, Mach 1
p__ · nozzle = de Laval = bell · pressure + heat → speed
p__ · exhaust velocity (v_e) · how fast the gas leaves
p__ · g_0 = standard gravity · 9.81 m/s^2, a constant
p__ · specific impulse (Isp) · $v_e \div g_0$, seconds, mileage → C13
p__ · regenerative cooling · fuel cools the walls first
p__ · solid rocket motor / grain · premixed block, can't shut off
p__ · booster · strap-on helper at liftoff (a job)
p__ · stage / staging · drop empty sections
p__ · thrust-to-weight ratio · thrust ÷ weight; > 1 lifts off

Also: on card C13 (specific impulse) add in the margin: **$Isp = v_e \div g_0$ · 1 kg gives 9.81 N for Isp seconds**.

SEC 4 · LEC 2 · CONCEPT CARDS

Route E: pages C18 and C19

Two notebook pages, two cards each, same six-line shorthand as C1–C17. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 8 — C18

EXHAUST VELOCITY (v_e) = ejection speed

= how fast the gas leaves the nozzle
WHY it is the engine's quality in one number: faster exhaust = more push from every kilo
\# kerosene 3.4 km/s · hydrogen 4.4 · ion engine 41 · chemical max ~4.5
≠ the ship's speed (v_e is the GAS)
→ Isp = $v_e \div g_0$ · $F = \dot{m} \times v_e$ · rocket equation

MASS FLOW RATE ($\dot{m}$) = m-dot

= kg of propellant thrown out per second
WHY thrust needs speed AND amount. $\dot{m}$ is the amount; it is also how fast the tanks empty
\# Merlin Vacuum 287 kg/s · F-1 2,600 kg/s · ion engine mg/s
≠ total propellant mass ($\dot{m}$ is per second)
→ $F = \dot{m} \times v_e$ · burn time = mass ÷ $\dot{m}$

WORD BANK · CONCEPT CARDS · scene 8 — C19

DE LAVAL NOZZLE = bell nozzle

= narrows to a throat, then widens into a bell
WHY chamber gas is hot and squeezed but slow. The nozzle trades pressure + heat for speed
\# Mach 1 at the throat → 3–4 km/s at the exit
≠ a garden-hose nozzle (subsonic only)
→ throat · expansion · exhaust velocity

BIPROPELLANT = fuel + oxidizer

= engine that burns two propellants together
WHY fire needs oxygen and space has none, so the rocket carries its own in a second tank
\# LH_2 + LOX 450 s · kerosene + LOX 350 s · hydrazine alone 225 s
≠ monopropellant (one liquid, no burn)
→ combustion chamber · Isp · hypergolic

C18: 10 + 11 lines of 22 · C19: 10 + 10 lines of 22

SEC 4 · LEC 2 · SAY IT OUT LOUD

Last page: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I will grade it like a friend would.

🗣 The friend's opening question

"So how does a rocket engine actually work? Isn't it just a big fire?"

✅ Three sentences you should be able to say

- It is a fire with only one way out. You pump fuel and oxygen into a chamber, they burn, and the hot squeezed gas has to leave through a narrow throat and a bell.
- The bell is the clever part: it turns pressure into speed, so the gas leaves at three or four kilometres a second.
- Two numbers describe it. Thrust is how much you throw per second times how fast, and specific impulse is the mileage. Chemical rockets push enormously hard and get poor mileage.

❓ Follow-ups a friend might ask

- "Why two tanks?" (no air in space, so it brings its own oxygen)
- "Why is the nozzle bell-shaped?" (the gas has to expand to speed up; past the speed of sound a widening pipe makes it faster)
- "Why is specific impulse in seconds?" (exhaust speed divided by 9.81; it is how long a kilo can hold up a kilo)
- "Why doesn't it melt?" (the cold fuel runs through the walls first)
- "Can't we just build a better chemical engine?" (no: the energy per kilo is fixed by chemistry, ceiling about 460 s)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Next: the rest of Section 4 (electric and nuclear engines). Send the transcript and slides when you have watched them.